

Arxiv Submission Guide

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Abstract—This module forms a critical component of the CEREBRUM framework, a community-structured graph attention engine for Knowledge Graph reasoning. It focuses on Framework Implementation Guide.

Index Terms—Knowledge Graph, Graph Attention, DSCF, CSA, Hierarchical Reasoning.

1 ARXIV SUBMISSION GUIDE

CEREBRUM — 16 Papers Ready for Submission

This guide covers the submission process, per-paper metadata, and checklist for submitting all sixteen CEREBRUM papers to arXiv.

1.1 arXiv Categories

All sixteen papers should be submitted to **primary category cs.IR** (Information Retrieval) with the following cross-list categories as appropriate:

Primary	Cross-list	Use for
cs.IR	cs.AI, cs.LG	All papers (default)
cs.AI	cs.IR, cs.NE	Papers 3, 4, 7 (neuromorphic / biological analog)
cs.LG	cs.IR, cs.AI	Papers 6, 15 (learning algorithms)
cs.DC	cs.IR, cs.AI	Papers 5, 13 (distributed / streaming)

1.2 Submission Order (Recommended)

Submit foundational papers first so cross-references resolve correctly.

Order	Paper	File	Category
1	DSCF/TSC Community Fusion	PAPER_001_DSCF_TSC.md	cs.AI
2	Community-Structured Attention	PAPER_002_CSA.md	cs.LG
3	Bridge Twin Nodes	PAPER_003_BRIDGE_TWIN.md	cs.NE
4	STD P Causal Edge Inference	PAPER_004_STD_P_CAUSAL.md	cs.NE
5	Holographic Federated Index	PAPER_005_HOLOGRAPHIC_INDEXING.md	cs.IR
6	Bayesian Beam Search	PAPER_006_BAYESIAN_BEAM.md	cs.IR
7	REM Cycle Maintenance	PAPER_007_REMCYCLE.md	cs.IR
8	Cross-Modal Signal Alignment	PAPER_008_CROSS_MODAL_ALIGNMENT.md	cs.LG
9	THALAMUS Ingestion Pipeline	PAPER_009_THALAMUS.md	cs.AI
10	Inference Validation	PAPER_010_INFERENCE_VALIDATION.md	cs.AI
11	Contradiction Materialization	PAPER_011_CONTRADICTION.md	cs.AI
12	Glass-Box Reasoning Studio	PAPER_012_GLASS_BOX_REASONING_STUDIO.md	cs.IR
13	Streaming Engine	PAPER_013_STREAMING_ENGINE.md	cs.IR
14	Metacognitive Verification	PAPER_014_METACOGNITIVE_VERIFICATION.md	cs.IR
15	Algorithmic Depth	PAPER_015_ALGORITHMIC_DEPTH.md	cs.IR
16	Production Hardening	PAPER_016_PRODUCTION_HARDENING.md	cs.IR

1.3 Per-Paper Metadata Template

Each submission requires the following metadata fields on arXiv:

Title: [From paper header]
Authors: Bryan Alexander Buchorn; Claude Sonnet 4.6 (Research Collaborator, Anthropic)
Abstract: [From paper Abstract section max 1,920 characters]
Comments: 16 pages. Part of the CEREBRUM framework series.

```
Code: https://github.com/[repo]
MSC-class: 68T30 (Knowledge representation)
ACM-class: I.2.4; H.3.3
License: CC BY 4.0 (arXiv default for non-
commercial research)
```

1.4 Formatting Requirements

arXiv accepts .tex (preferred) and .pdf. Our papers are in Markdown; conversion is required.

1.4.1 Markdown → LaTeX conversion

```
pip install pandoc
pandoc docs/arxiv/PAPER_001_DSCF_TSC.md -o
  arxiv_submission/paper_001.tex \
  --template arxiv_template.tex \
  --bibliography refs.bib
```

A LaTeX template (arxiv_template.tex) following arXiv's article document class should be prepared once and reused for all sixteen papers.

1.4.2 Required LaTeX elements

- \usepackageamsmath — for displayed equations
- \usepackagebooktabs — for benchmark tables
- \usepackagehyperref — for cross-references and URLs
- All figures as .pdf or .png at 300 DPI minimum

1.4.3 Figure requirements

Each paper should include at minimum:

- 1) An architecture diagram (flowchart of the algorithm)
- 2) A results table (benchmark comparison)

Figures should be prepared in a vector format (PDF) for crisp rendering.

1.5 Abstract Length Check

arXiv abstracts are capped at 1,920 characters. Check each abstract:

```
for paper in docs/arxiv/PAPER_*.md; do
  abstract=$(awk '/^### Abstract/{found=1;
    next} found && /^###/{exit} found{print
    }' "$paper")
  count=$(echo "$abstract" | wc -c)
  echo "$paper: $count chars"
done
```

1.6 Pre-Submission Checklist (per paper)

- Abstract $\leq$ 1,920 characters
- All equations render correctly in LaTeX
- All tables have \toprule/\midrule/\bottomrule (booktabs style)
- All citations in refs.bib with correct BibTeX keys
- Author affiliations: "Independent Researcher" + "Anthropic"

Prior art section explicitly differentiates from closest prior work

Benchmark numbers match release/BENCHMARKS.md

GitHub/code link included in Comments field

License selected: CC BY 4.0

Cross-list categories correct for paper topic

1.7 Citation Cross-Referencing

Papers cite each other. Use consistent arXiv IDs once papers 1–4 are submitted. Placeholder format until IDs are assigned:

```
@article{buchorn2026dscf,
  title = {Dual-Signal Community Fusion for
    Knowledge Graph Attention},
  author = {Buchorn, Bryan Alexander and {
    Claude Sonnet 4.6}},
  journal = {arXiv preprint arXiv:2026.XXXXX},
  year = {2026}
}
```

1.8 Comprehensive Paper (Parallax_White_Paper_arXiv.md)

The comprehensive umbrella paper (docs/Parallax_White_Paper_arXiv.md) covers all subsystems in one document and should be submitted to cs.IR as the primary reference after all sixteen individual papers are live. Title it:

>CEREBRUM: Community-Structured Graph Attention for Zero-Shot Multi-Hop Knowledge Graph Reasoning

This paper should cite all sixteen subsystem papers by their arXiv IDs.

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REFERENCES